(ix) $\sec y = 2\sin x + C$ (x) $\frac{1}{2}\left[(x+y)+\sin(x+y)\cos(x+y)\right] = x + C$

Exercise 10.6

1. $\sin\left(\frac{y}{x}\right) = \log|Cx|$ 2. $y = Ce^{\frac{x^3}{3y^3}}$ 3. $e^{\frac{x}{y}} = \log|Cy|$ 4. $3x^2y + 2y^3 = C$
5. $xy^2 - x^2y = C$ 6. $C = xe^{\tan\left(\frac{y}{x}\right)}$ 7. $y + 3xe^{\frac{y}{x}} = 3$ 8. $x_0 = \pm\sqrt{3}e$

Exercise 10.7

1. $y = \sin x + C\cos x$ 2. $y = \frac{\sin^{-1}x}{\sqrt{1-x^2}} + C\left(1-x^2\right)^{-\frac{1}{2}}$ 3. $(y+\cos x)x = \sin x + C$
4. $y\left(x^2+1\right) = \frac{x}{2}\sqrt{x^2+4} + \frac{1}{2}\log\left|x+\sqrt{x^2+4}\right| + C$ 5. $xy^2 = 2y^5 + C$
6. $xy\sin x + \cos x = C$ 7. $ye^{\sin^{-1}x} = \frac{e^{2\sin^{-1}x}}{2} + C$ 8. $y\left(\frac{1+\sqrt{x}}{1-\sqrt{x}}\right) = x + \frac{2}{3}x\sqrt{x} + C$
9. $xy + \tan^{-1}y = C$ 10. $y\log x + \frac{\cos 2x}{2} = C$ 11. $2y = (x+a)^4 + 2C(x+a)^2$
12. $y\left(1+x^3\right) = \frac{x}{2} - \frac{\sin 2x}{4} + C$ 13. $4yx = 2x^2\log x - x^2 + 4C$ 14. $x^2y = \frac{x^4}{4}\log x - \frac{x^4}{16} + C$
15. $2x^3y = x^2 + 3$

Exercise 10.8

1. After 10 hours the number of bacteria as 9 times the original number of bacteria.
2. $P = 300000\left(\frac{4}{3}\right)^{\frac{t}{40}}$ 3. $i = Ce^{-\frac{Rt}{L}}$ 4. $v = \frac{10}{e^2}$ 5. $P = 10000e^{0.075}$
6. $\frac{9^{10}}{10^8}$ % of the radioactive element will remain after 1000 years.
7. (i) $65.33°C$ (ii) 51.91 *mts*
8. (i) $T \simeq 151°F$ (ii) $t = 22.523$. She drunk the coffee between 10.22 and 10.30 approximately.
9. $20°$ 10. $x = 100\left(1-e^{\frac{-3t}{50}}\right)$

Exercise 10.9

Q	1	2	3	4	5	6	7	8	9	10
A	(1)	(2)	(3)	(2)	(2)	(3)	(3)	(2)	(2)	(3)
Q	11	12	13	14	15	16	17	18	19	20
A	(3)	(3)	(1)	(1)	(2)	(3)	(2)	(4)	(2)	(4)
Q	21	22	23	24	25					
A	(1)	(1)	(2)	(2)	(1)					

Chapter 11

EXERCISE 11.1

(1)

Values of Random Variable	0	1	2	3	Total
Number of points in inverse image	1	3	3	1	8

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